

TEAM MEMBERS

1. Akinola Summayah
2. Daniyan Titilayo
3. Goodness Muoka
4. Opeyemi Yahya

Selected Project: **Campus Resource Center Borrowing System**

Class being Tested: **Resources**

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output/behaviour
TC1	mark_borrowed	Borrow an available resource	Is available = True	Is_available becomes False
TC2	mark_borrowed	Return a borrowed resource	Is_available = False	Is_available becomes True
TC3	mark_borrowed	Try to borrow a resource that's already borrowed (edge case)	Is_available = False	Borrow is rejected , is_available stays False, and message shown
TC4	init	Create a resource with an empty ID (invalid)	Resource_id = ""	Resource creation rejected/error shown
TC5	init	Create a resource with an empty name (invalid)	name = ""	Resource creation rejected or error shown

Class being Tested: **Student**

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output/behaviour
TC1	init	Register a new student with valid ID and name	Student_id = "S001", name = "Amaka"	Student object created successful
TC2	add_borrowed_resource	Add a resource to a student's borrowed list	Resource_id = "R001"	R001 appears in borrowed_resource list

TC3	remove_borrowed_resource	Try to remove a resource that isn't in the student's list (edge case)	Resource_id = "R888" (not borrowed)	Handled well, no crash, list unchanged
TC4	init	Register a student with an empty name (invalid)	Name = ""	Student creation rejected or error shown
TC5	init	Register a student with a duplicate student_id (invalid)	Student_id = an ID already used	Registration rejected or error shown.

Class being Tested: **Borrowing**

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output/behaviour
TC1	init	Create a new borrowing record with valid details	Borrowing_id = "B001", Student_id = "S001", Resource_id = "R001", Borrow_date = "2026-09-01"	Borrowing object created successfully, status set to "Borrowed"
TC2	mark_returned	Mark a borrowed item as returned	return_date = "2026-09-07"	Status changes to "Returned", return_date is set to "2026-09-07"
TC3	mark_returned	Try to mark an already-returned record as returned again (edge case)	return_date = "2026-09-15"	Handled well, no crash, status/return_date unchanged
TC4	init	Create a borrowing record with an empty student_id (invalid)	student_id = ""	Borrowing creation rejected and error message shown
TC5	init	Create a borrowing record with an empty resource_id (invalid)	resource_id = ""	Borrowing creation rejected and error message shown